

MA3236 NONLINEAR PROGRAMMING

Semester 1, AY 2026/2027

Tutorial 5

Review: Lecture 10 — Steepest descent

Let $f : \mathbf{R}^n \rightarrow \mathbf{R}$, $\mathbf{g}^{(k)} = \nabla f(\mathbf{x}^{(k)})$, and $\mathbf{d}^{(k)} = -\mathbf{g}^{(k)}$. For $\mathbf{g}^{(k)} \neq \mathbf{0}$, the direction is descending and a line-search step is

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}, \quad \alpha^{(k)} > 0.$$

- **Smooth quadratic upper model.** If ∇f is L -Lipschitz, then

$$f(\mathbf{y}) \leq M_L(\mathbf{y}; \mathbf{x}) := f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) + \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2.$$

The model minimizer is $\mathbf{x} - L^{-1} \nabla f(\mathbf{x})$, and

$$f\left(\mathbf{x} - \frac{1}{L} \nabla f(\mathbf{x})\right) \leq f(\mathbf{x}) - \frac{1}{2L} \|\nabla f(\mathbf{x})\|^2.$$

Thus $\alpha = 1/L$ satisfies the sufficient-decrease test in Armijo backtracking line search with $\sigma = 1/2$; this step bound is sufficient, not necessary.

- **Exact search on a positive-definite quadratic.** For $q(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} - \mathbf{c}^T \mathbf{x}$, with symmetric $\mathbf{Q} \succ 0$, the SD direction is $\mathbf{d} = -\mathbf{g}$ and

$$\alpha_{\text{exact}} = \frac{\|\mathbf{g}\|^2}{\mathbf{g}^T \mathbf{Q} \mathbf{g}}.$$

Successive gradients are orthogonal in exact arithmetic.

- **Rate and preconditioning.** If f is μ -strongly convex and L -smooth, the $1/L$ -step SD method satisfies

$$E^{(k+1)} \leq \left(1 - \frac{1}{\kappa}\right) E^{(k)},$$

where $E^{(k)} = f(\mathbf{x}^{(k)}) - f^*$ and $\kappa = L/\mu$. For a quadratic, an exact-search SD worst-case gap factor is $((\kappa - 1)/(\kappa + 1))^2$. A symmetric PD preconditioner $\mathbf{B}$ uses $\mathbf{d} = -\mathbf{B}^{-1} \mathbf{g}$; its effective condition number is that of $\mathbf{B}^{-1/2} \mathbf{Q} \mathbf{B}^{-1/2}$.

Review: Lecture 11 — Conjugate directions and CG

For the same quadratic, write $\mathbf{g}^{(k)} = \mathbf{Q}\mathbf{x}^{(k)} - \mathbf{c}$. Directions $\mathbf{d}^{(i)}$ and $\mathbf{d}^{(j)}$ are $\mathbf{Q}$ -conjugate when $(\mathbf{d}^{(i)})^T \mathbf{Q} \mathbf{d}^{(j)} = 0$ for $i \neq j$. Exact line searches along conjugate directions preserve all earlier line-search optimality conditions:

$$\alpha^{(k)} = -\frac{(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)}}{(\mathbf{d}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k)}}.$$

After $k+1$ independent conjugate searches, the iterate minimizes q over the affine subspace

$$\mathbf{x}^{(0)} + \text{span}\{\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(k)}\}.$$

- **Conjugate gradient (CG).** It constructs the conjugate directions recursively:

$$\begin{aligned} \mathbf{d}^{(0)} &= -\mathbf{g}^{(0)}, & \alpha^{(k)} &= \frac{\|\mathbf{g}^{(k)}\|^2}{(\mathbf{d}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k)}}, \\ \mathbf{x}^{(k+1)} &= \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}, & \mathbf{g}^{(k+1)} &= \mathbf{g}^{(k)} + \alpha^{(k)} \mathbf{Q} \mathbf{d}^{(k)}, \\ \beta^{(k+1)} &= \frac{\|\mathbf{g}^{(k+1)}\|^2}{\|\mathbf{g}^{(k)}\|^2}, & \mathbf{d}^{(k+1)} &= -\mathbf{g}^{(k+1)} + \beta^{(k+1)} \mathbf{d}^{(k)}. \end{aligned}$$

In exact arithmetic, CG terminates in at most n updates; fewer can suffice when the initial error involves fewer distinct eigenvalues. Finite precision can spoil exact conjugacy and termination.

- **Preconditioned CG (PCG).** For a fixed symmetric PD $\mathbf{B}$, set $\mathbf{z}^{(k)} = \mathbf{B}^{-1} \mathbf{g}^{(k)}$ and start with $\mathbf{d}^{(0)} = -\mathbf{z}^{(0)}$. Its first step is

$$\alpha^{(0)} = \frac{(\mathbf{g}^{(0)})^T \mathbf{z}^{(0)}}{(\mathbf{d}^{(0)})^T \mathbf{Q} \mathbf{d}^{(0)}}.$$

PCG is CG applied in transformed coordinates to $\mathbf{B}^{-1/2} \mathbf{Q} \mathbf{B}^{-1/2}$. A better effective condition number can reduce iterations, but applying $\mathbf{B}^{-1}$ also has a cost.

1. **A smooth upper model and one gradient step.** Let

$$q(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x}, \quad \mathbf{Q} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{x}^{(0)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

- (a) Find the eigenvalues of $\mathbf{Q}$, and hence valid sharp L -smoothness and strong-convexity constants. At $\mathbf{x}^{(0)}$, write the quadratic upper model $M_L(\mathbf{y}; \mathbf{x}^{(0)})$, prove that it majorizes q , and find its minimizer.
- (b) Compute the resulting SD iterate $\mathbf{x}^{(1)}$ and both function values.
- (c) Check the Armijo condition with $\sigma = 1/2$ and $\alpha = 1/L$. Compare the actual gap ratio with the $1 - \mu/L$ bound.

2. **Exact-search SD, a rate, and preconditioning.** Consider

$$q(\mathbf{x}) = \frac{1}{2}(x_1^2 + 4x_2^2), \quad \mathbf{x}^{(0)} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}.$$

- (a) Compute $\alpha^{(0)}, \mathbf{x}^{(1)}, \alpha^{(1)}$, and $\mathbf{x}^{(2)}$ under exact-line-search SD. Verify that $(\mathbf{g}^{(0)})^T \mathbf{g}^{(1)} = 0$.
- (b) Prove by induction that

$$\alpha^{(k)} = \frac{2}{5}, \quad \mathbf{x}^{(k)} = \left(\frac{3}{5}\right)^k \begin{pmatrix} 4 \\ (-1)^k \end{pmatrix}.$$

Also show that

$$\frac{q(\mathbf{x}^{(k)})}{q(\mathbf{x}^{(0)})} = \left(\frac{9}{25}\right)^k.$$

Relate the factor to the condition number of $\mathbf{Q}$.

- (c) Find the first k for which $q(\mathbf{x}^{(k)})/q(\mathbf{x}^{(0)}) \leq 10^{-3}$. What happens if preconditioned SD uses $\mathbf{B} = \text{diag}(1, 4)$ with exact line search?

3. **Constructing two conjugate directions.** Let

$$q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} - \mathbf{c}^T\mathbf{x}, \quad \mathbf{Q} = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Start from

$$\mathbf{x}^{(0)} = \mathbf{0}, \quad \mathbf{d}^{(0)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

- (a) Verify that $\mathbf{Q} \succ 0$. Find the unique direction of the form $\mathbf{d}^{(1)} = (t, 1)^T$ that is $\mathbf{Q}$ -conjugate to $\mathbf{d}^{(0)}$.
- (b) Perform exact line searches along $\mathbf{d}^{(0)}$ and then $\mathbf{d}^{(1)}$. Give both step lengths, both iterates, and the final gradient.
- (c) Explain using gradient orthogonality why the second step does not undo the first line minimization, and identify the affine subspace minimized over after both steps.

4. **CG versus preconditioned CG.** For $\frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} - \mathbf{c}^T\mathbf{x}$, use

$$\mathbf{Q} = \text{diag}(1, 4, 9), \quad \mathbf{c} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{x}^{(0)} = \mathbf{0}, \quad \mathbf{B} = \text{diag}(1, 4, 3).$$

- (a) Find $\kappa(\mathbf{Q})$ and the effective condition number $\kappa(\mathbf{B}^{-1/2}\mathbf{Q}\mathbf{B}^{-1/2})$.
- (b) Compute the first direction, exact step length, and iterate for ordinary CG and for PCG with $\mathbf{B}$.
- (c) In exact arithmetic, how many updates does each method need to reach the minimizer? Justify your answer from the active eigenvalues, and state one reason the conclusion can fail in floating-point arithmetic.